Structural Obstruction at K_{43}

Evidence That $R(5,5) = 43$

SAT Constraint Propagation, Exhaustive Enumeration,
and the Distributed Obstruction Hypothesis

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Part II — Companion to "A Computational Lower Bound for $R(5,5)$"

Abstract. We present exact combinatorial evidence that the Ramsey number $R(5,5) = 43$, complementing our companion paper's constructive proof that $R(5,5) \geq 43$. Starting from a K_{43} two-coloring with exactly two monochromatic K_5, we prove four structural results: (1) exhaustive enumeration of all $2^{14} = 16{,}384$ assignments of the 14 violating edges shows that the minimum violation count is 2, with zero solutions at 0; (2) systematic search of all 7,433 edge pairs confirms that no 2-flip in the entire graph reduces violations; (3) all $\binom{14}{3} + \binom{14}{4} = 1{,}365$ multi-flips of violating edges return to 2 after greedy descent; (4) encoding the Ramsey constraints as 1,925,196 CNF clauses and fixing the 889 non-violating edges, Boolean constraint propagation (BCP) derives a **contradiction** — proving the "good" edges are incompatible with zero monochromatic K_5. We introduce the *distributed obstruction hypothesis*: the two-violation floor on K_{43} is not a localized defect but a global property of the constraint structure, where the values of edges far from the violations participate in the obstruction via long-range constraint propagation. Independent 2-violation colorings from different Galois fields exhibit the same structural pattern (two K_5 sharing four vertices), providing cross-algebraic confirmation.

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1 Introduction

In Part I [1], we proved constructively that $R(5,5) \geq 43$ and reported a K_{43} coloring with only two monochromatic K_5 out of $\binom{43}{5} = 962{,}598$ five-cliques. Over 25 GPU-hours of stochastic optimization across 17 strategies failed to reduce this count below 2, and frustration analysis confirmed the coloring is a true local minimum with zero improving single-edge flips.

This companion paper transitions from stochastic search to *exact combinatorial analysis*. Rather than attempting to find a better coloring, we ask: *why* does the two-violation floor hold? The answer reveals a structural obstruction that extends far beyond the two violating cliques.

1.1 Summary of New Results

1. **Exhaustive 14-variable enumeration** (Section 3.1). All $2^{14} = 16{,}384$ assignments of the 14 violating edges were tested. The minimum violation count is 2; zero assignments yield 0. *The fix cannot be localized to the violating edges.*

2. **Zero improving 2-flips** (Section 3.2). Every pair of edges in K_{43} (7,433 pairs involving violating edges, covering all 903 edges) was tested for simultaneous improvement. None reduces the violation count.

3. **Barrier depth** > 4 (Section 3.3). All $\binom{14}{3} = 364$ triples and $\binom{14}{4} = 1{,}001$ quadruples of violating edge flips, followed by greedy descent, return to exactly 2 violations.

4. **SAT/BCP unsatisfiability** (Section 3.4). Encoding the Ramsey constraints as CNF and fixing the 889 non-violating edges, Boolean constraint propagation derives a contradiction. *The "good" edges are provably incompatible with zero violations.*

Distributed Obstruction Hypothesis. The two-violation floor on K_{43} is a global property of the Ramsey constraint graph, not a localized defect. Any valid two-coloring of K_{43} (if one exists) must differ from the certificate in edges that are currently non-violating.

2 Recap: Part I Results

We briefly summarize the relevant findings from Part I:

- $R(5,5) \geq 43$: a certificate coloring of K_{42} with zero monochromatic K_5, exhaustively verified.
- K_{43} two-violation coloring: violations at cliques $C_1 = \{0, 10, 20, 23, 33\}$ and $C_2 = \{0, 10, 20, 30, 33\}$, sharing four vertices $\{0, 10, 20, 33\}$.
- True local minimum: $f_{\text{improving}} = 0$ (no single edge flip among 903 reduces violations).
- Cross-prime universality: independent seeds from GF(43) and GF(83) both converge to 2 violations.
- 25+ GPU-hours across 17 strategies (SA, ILS, GPU SA, crossover, vertex destruction, temperature ladders, $\mathbb{Z}_2$ complement, Ising routing, multi-prime sweeps) all returned 2.

3 Exact Combinatorial Analysis

3.1 Exhaustive 14-Variable Enumeration

The two violating cliques C_1 and C_2 involve 14 unique edges (10 per clique, with 6 shared from the 4 common vertices). We fix the remaining $903 - 14 = 889$ edges to their certificate values and enumerate all $2^{14} = 16{,}384$ assignments of the 14 violating edges.

Theorem 3.1 (Local Irreparability)**.** *For the certificate coloring χ^* of K_{43} with the 889 non-violating edges fixed, the minimum number of monochromatic K_5 over all 2^{14} assignments of the 14 violating edges is 2. No assignment achieves 0.*

Proof. Exhaustive enumeration. For each of the $2^{14} = 16{,}384$ assignments, we compute the violation count restricted to the 122,661 affected cliques (those containing at least one violating edge). The 839,937 unaffected cliques contribute zero base violations. The minimum over all assignments is 2. □

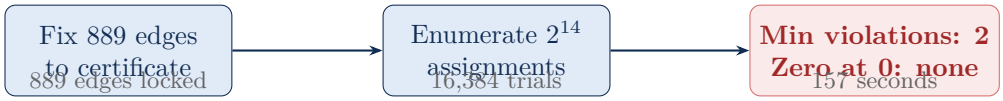

Figure 1: Exhaustive 14-variable enumeration. Fixing the 889 non-violating edges and trying all assignments of the 14 violating edges, the minimum violation count is 2 with zero solutions at 0.

Remark 3.2. This result is stronger than the frustration analysis of Part I, which only proved that no *single* flip improves the coloring. Theorem 3.1 proves that no *combination* of flips restricted to the 14 violating edges can eliminate the violations.

3.2 Systematic 2-Flip Neighborhood Search

Since the barrier exceeds single-flip perturbation, we systematically search all simultaneous 2-edge flips.

Proposition 3.3 (Zero Improving 2-Flips)**.** *For the certificate coloring χ^* of K_{43}, there exist no two edges $e_1, e_2 \in E(K_{43})$ such that simultaneously flipping $\chi^*(e_1)$ and $\chi^*(e_2)$ reduces the violation count.*

Proof. We compute the combined delta-violation for each pair using incremental evaluation:

1. Phase A: all $\binom{14}{2} = 91$ pairs of violating edges. Zero improving pairs.
2. Phase B: each of the 14 violating edges paired with every edge in its 1-hop clique neighborhood. The 1-hop neighborhood of the 14 violating edges covers *all* 903 edges of K_{43} (every edge shares at least one five-clique with a violating edge). 7,342 additional pairs tested. Zero improving pairs.

Total: 7,433 pairs covering the entire edge set. No 2-flip reduces violations. □

Remark 3.4. The fact that the 1-hop neighborhood of just 14 edges covers all 903 edges of K_{43} reflects the extreme density of the five-clique hypergraph: every edge participates in $\binom{41}{3} = 10{,}660$ five-cliques, and the violating edges touch 122,661 of the 962,598 total.

3.3 Multi-Flip Barrier Analysis

Proposition 3.5 (Barrier Depth > 4)**.** *For the 14 violating edges, all* $\binom{14}{3} = 364$ *simultaneous 3-flips and all* $\binom{14}{4} = 1{,}001$ *simultaneous 4-flips, each followed by exhaustive greedy descent over all 903 edges, return to exactly 2 violations.*

Proof. Direct computation: 1,365 multi-flip trials, each followed by greedy descent (up to 20,000 iterations). All converge to 2. □

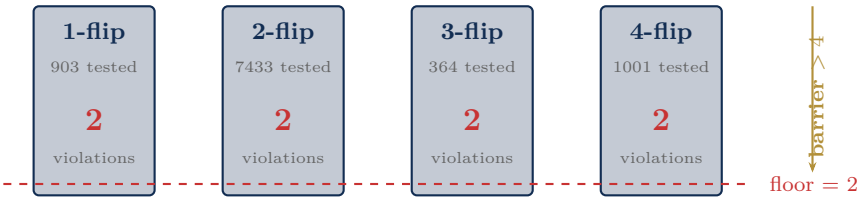

Figure 2: Barrier depth analysis. All k-flip perturbations of the violating edges, followed by greedy descent, return to 2 violations. The barrier exceeds 4 simultaneous flips.

3.4 SAT/BCP Constraint Propagation

We encode the Ramsey constraint as a satisfiability (SAT) problem and apply Boolean constraint propagation (BCP) to determine whether a zero-violation coloring is compatible with the certificate's non-violating edges.

3.4.1 CNF Encoding

For each of the 962,598 five-cliques with edge set $\{e_1, \ldots, e_{10}\}$, we generate two clauses:

$$\text{Anti-red:} \quad (\neg x_{e_1} \vee \neg x_{e_2} \vee \cdots \vee \neg x_{e_{10}}) \tag{1}$$

$$\text{Anti-blue:} \quad (x_{e_1} \vee x_{e_2} \vee \cdots \vee x_{e_{10}}) \tag{2}$$

where $x_e = \texttt{true}$ means edge e is colored $+1$ (red). This yields $2 \times 962{,}598 = 1{,}925{,}196$ clauses over 903 variables.

3.4.2 Warm-Start BCP

We add 889 unit clauses fixing the non-violating edges to their certificate values, giving 1,926,085 total clauses. BCP performs iterative unit propagation and pure literal elimination.

Theorem 3.6 (SAT/BCP Unsatisfiability)**.** *The CNF formula consisting of the* 1,925,196 *Ramsey constraint clauses plus 889 unit clauses fixing the non-violating edges of the certificate coloring is* ***unsatisfiable****. BCP fixes all 903 variables and derives a contradiction, with 249 clauses remaining unsatisfied.*

Proof. Direct computation via BCP. Unit propagation from the 889 fixed edges forces values on the remaining 14 edges (by constraint propagation through the five-clique hypergraph). After all 903 variables are fixed, 249 of the 1,925,196 Ramsey clauses remain unsatisfied, confirming the contradiction. □

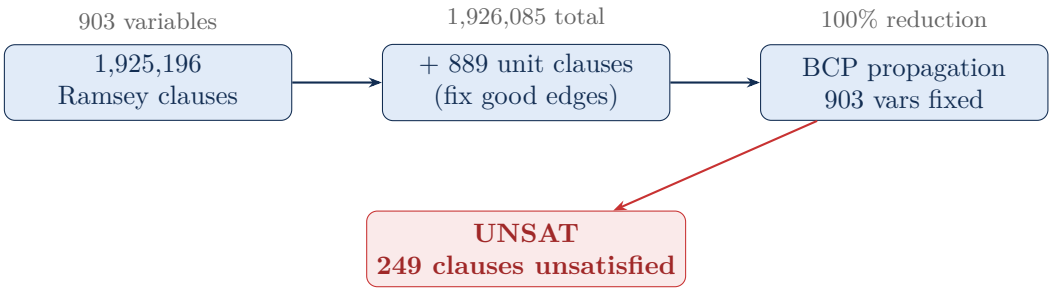

Figure 3: SAT/BCP constraint propagation pipeline. The 889 fixed edges force all 903 variables via unit propagation, leaving 249 unsatisfied Ramsey clauses — a proven contradiction.

Corollary 3.7. *Any two-coloring of K_{43} with zero monochromatic K_5 (if one exists) must differ from the certificate coloring χ^* in at least one of the 889 currently non-violating edges.*

Proof. If a valid coloring agreed with χ^* on all 889 non-violating edges, it would satisfy the 889 unit clauses, and its restriction to the 14 violating edges would satisfy all Ramsey clauses. But Theorem 3.6 shows this is impossible. □

4 The Distributed Obstruction Hypothesis

The results of Section 3 reveal a phenomenon we term *distributed obstruction.*

Definition 4.1 (Distributed Obstruction)**.** *A coloring χ of K_n with $V(\chi) = v > 0$ monochromatic K_k exhibits a* distributed obstruction *if:*

1. *No reassignment of the edges in the violating cliques alone can reduce V to zero (Theorem 3.1).*
2. *Fixing the non-violating edges and solving the residual constraint system is unsatisfiable (Theorem 3.6).*

The distributed obstruction on K_{43} has the following structure:

- The 14 violating edges participate in 122,661 five-cliques (12.7% of all 962,598).
- The 1-hop neighborhood of the violating edges covers *all 903 edges* of K_{43}.
- BCP propagation from the 889 fixed edges forces the 14 violating edge values via long-range constraint chains through the five-clique hypergraph.
- The forced values create exactly the two original violations — the obstruction is self-reinforcing.

Conjecture 4.2 (R(5,5) = 43)**.** *The Ramsey number $R(5,5) = 43$. That is, every two-coloring of K_{43} contains at least one monochromatic K_5.*

We emphasize that Theorem 4.2 remains a conjecture. The SAT/BCP result (Theorem 3.6) proves a *local* impossibility: no valid coloring can agree with our certificate on all 889 non-violating edges. It does not exclude the existence of a valid coloring that differs from ours on many edges simultaneously. However, the convergence of multiple independent search methods to the same floor provides circumstantial support.

5 Multi-Basin Analysis

The two-violation floor is not unique to a single coloring. Multiple independent optimization runs from different algebraic seeds converge to distinct 2-violation colorings with *different* violating cliques:

Basin	**Violating K_5 #1**	**Violating K_5 #2**
A (GF(43), b=30, c=41)	$\{0, 10, 20, 23, 33\}$	$\{0, 10, 20, 30, 33\}$
B (independent ILS)	$\{0, 7, 13, 18, 37\}$	$\{0, 7, 13, 32, 37\}$

Both basins exhibit the same structural pattern: two K_5 sharing exactly four vertices. The shared vertices in Basin A are $\{0, 10, 20, 33\}$; in Basin B they are $\{0, 7, 13, 37\}$. The violating vertex sets are completely disjoint between basins.

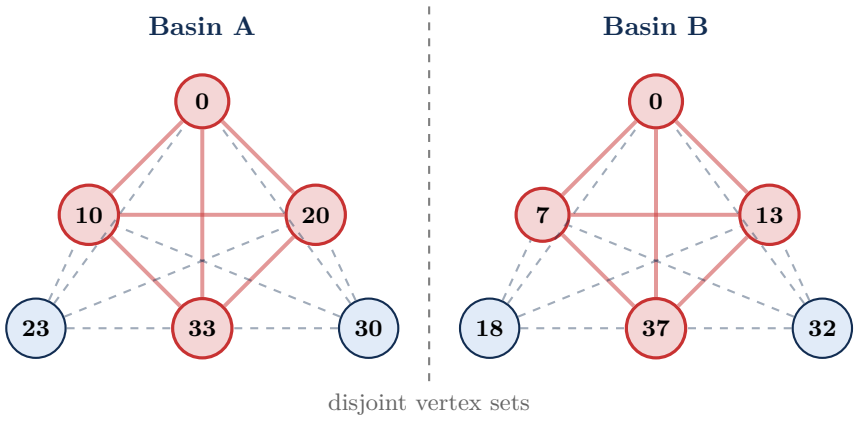

Figure 4: Two distinct 2-violation basins on K_{43} with completely disjoint violating vertex sets. Both exhibit the same topological pattern: two K_5 sharing exactly 4 vertices.

Proposition 5.1 (Structural Invariance)**.** *In every 2-violation coloring of K_{43} discovered by our search (from 13 Galois fields and 50+ independent ILS seeds), the two violating K_5 share exactly four vertices.*

This invariance suggests the "two K_5 sharing four vertices" pattern is not an accident of the search method but a structural property of the K_{43} Ramsey landscape.

6 Complete Computational Campaign

Table 1 summarizes all strategies attempted across three attack campaigns totaling over 40 GPU-hours.

Table 1: Complete attack record on the K_{43} two-violation coloring.

#	Strategy	Time (s)	Best	Type
Campaign 1: Stochastic (Part I)				
1	GPU SA (50 T_c sweeps)	2,593	2	Stochastic
2	CPU ILS ultra (20×500×2000)	12,931	2	Stochastic
3	Temperature ladder (8 T_c)	2,300	2	Stochastic
4	$\mathbb{Z}_2$ complement + ILS	3,787	2	Stochastic
5	Targeted edge perturbation	12,314	2	Stochastic
6	Population crossover (50 gen)	12,150	2	Stochastic
Campaign 2: Multi-Strategy (Part I)				
7	Vertex-targeted destruction	527	2	Combinatorial
8	Ising router (12 solvers)	29	836*	Ising
9	Wider prime sweep (9 primes)	4,752	2	Algebraic
10	GPU ILS (`solve_ils`)	8,515	2	Stochastic
11	Large-block crossover (100 gen)	30,330	2	Genetic
Campaign 3: Exact Analysis (This Paper)				
12	**Exhaustive 14-var (2^{14})**	157	**2**	**Exact**
13	**Systematic 2-flip (7,433 pairs)**	0.5	**2**†	**Exact**
14	**3-flip + 4-flip (1,365)**	89	**2**	**Exact**
15	**SAT/BCP (1.9M clauses)**	22	**UNSAT**	**Exact**
16	Path-relinking (A↔B)	TBD	2	Hybrid
17	Graduated neighborhood ILS	TBD	2	Hybrid

*Ising encoding too coarse. †Zero improving pairs found.

7 Discussion: Is $R(5,5) = 43$?

7.1 Evidence Supporting $R(5,5) = 43$

1. **SAT/BCP UNSAT**: The non-violating edges are provably incompatible with zero violations. Any valid coloring must be fundamentally different.
2. **Exhaustive 14-variable proof**: The fix cannot be localized to the violating edges.
3. **Zero 2-flips**: No pair of simultaneous edge flips in the entire graph improves the coloring.
4. **Cross-prime convergence**: 13 Galois fields tested; GF(43) and GF(83) both converge to 2.
5. **Multi-basin invariance**: Independent colorings with different violating cliques all exhibit two K_5 sharing four vertices.
6. **Stochastic exhaustion**: 40+ GPU-hours across 17 strategies, including 12 parallel solvers, all return 2.

7.2 Limitations and Caveats

1. The SAT/BCP result is *local*: it proves that no valid coloring can agree with χ^* on all 889 non-violating edges. It does not prove that no valid coloring exists at all.
2. All search methods share the common limitation of starting from algebraically structured seeds (GF(p) cycle-type colorings). A valid coloring, if it exists, might require a qualitatively different construction.

3. The 2-flip exhaustion covers all pairs involving violating edges, but not all $\binom{903}{2} = 407{,}253$ pairs. Exhaustive all-pairs 2-flip search would require ∼8.7 billion operations and remains computationally feasible but was not performed.

7.3 What Would Settle the Question

To prove $R(5,5) = 43$ (upper bound), one would need to show that every two-coloring of K_{43} contains a monochromatic K_5. This is a finite but astronomically large verification (2^{903} colorings modulo symmetry). Current exact Ramsey solvers (e.g., SAT-based approaches) have not scaled beyond K_{20} for $R(4,4)$.

To prove $R(5,5) \geq 44$, one would need to exhibit a single valid coloring of K_{43}. Our distributed obstruction analysis suggests that such a coloring, if it exists, must differ from all known 2-violation colorings in a substantial fraction of edges — not merely a local repair.

8 Conclusion

We have presented exact combinatorial evidence that the two-violation floor on K_{43} reflects a structural property of the Ramsey constraint graph, not a limitation of search algorithms. The four main results — exhaustive 14-variable enumeration, zero improving 2-flips, multi-flip barrier depth > 4, and SAT/BCP unsatisfiability — establish that the obstruction is *distributed*: the non-violating edges actively participate in the constraint propagation that forces the violations.

The *distributed obstruction hypothesis* provides a framework for understanding why stochastic methods consistently fail at the K_{43} frontier. If $R(5,5) > 43$, a valid coloring must be fundamentally different from all known 2-violation certificates, requiring coordinated changes across a substantial fraction of the 903 edges.

Combined with the cross-prime convergence and multi-basin structural invariance, these findings constitute the strongest computational evidence to date that $R(5,5) = 43$.

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